

PREDICTING STUDENT ACADEMIC PERFORMANCE:

A Machine Learning Approach for Early-Stage Risk Prediction Without Grade Leakage

A Project Report

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Department of Computer Science and Engineering

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Table of Contents

Table of Contents.....	2
Chapter 1: Introduction	3
1.1 Background	3
1.2 Problem Statement.....	3
1.3 Objectives.....	3
1.4 Scope.....	4
1.5 Motivation.....	4
1.6 Applications.....	4
1.7 Expected Outcomes	4
Chapter 2: Literature Review	6
2.1 Existing Approaches	6
2.2 Research Studies	6
2.3 Research Gap	6
Chapter 3: Methodology.....	8
3.1 Dataset Description.....	8
3.2 Data Preprocessing	8
3.3 Algorithms Used.....	8
Chapter 4: Implementation	10
4.1 Exploratory Data Analysis (EDA)	10
4.2 Feature Engineering.....	10
4.3 Model Development	10
Chapter 5: Results and Discussion	12
5.1 Performance Metrics	12
5.2 Comparative Analysis.....	12
Chapter 6: Conclusion and Future Scope.....	13
6.1 Conclusion.....	13
6.2 Future Scope	13
References	14

Chapter 1: Introduction

1.1 Background

Modern education is producing more data than institutions know what to do with. Learning management systems, attendance registers, and administrative records now capture a continuous stream of information about how students engage with their coursework — yet very little of this data is turned into insight that actually helps a student before it is too late. This gap is what the field of Educational Data Mining (EDM) sets out to close: applying data analysis techniques to large educational datasets in order to surface the patterns in behaviour, study habits, and personal circumstances that shape academic outcomes.

Traditionally, institutions have leaned on reactive tools — end-of-semester grades, final exam results — to spot students who are struggling. The problem with this approach is timing. A poor final grade confirms that a student needed help, but it arrives after the window for meaningful intervention has already closed. What is needed instead is a shift from hindsight to foresight: predictive systems that flag risk early enough for a teacher or counsellor to actually do something about it.

1.2 Problem Statement

A common shortcut in academic performance prediction is to use mid-term grades (G1 and G2) as inputs for forecasting the final grade (G3). This tends to produce very accurate models, but it is a somewhat hollow accuracy — by the time G1 and G2 exist, the semester is already well underway, and the predictive exercise stops being 'early warning' in any meaningful sense. This is a form of data leakage: the model is being handed information that would not realistically be available at the moment an intervention needs to happen.

This project sets out to solve a stricter, more useful version of the problem: build a machine learning model that predicts a student's final performance using only behavioural, demographic, and lifestyle data — deliberately excluding G1 and G2. The result is a model that can genuinely be used at the very start of a term, giving educators a real head start rather than a confirmation of what they could already see coming.

1.3 Objectives

- Develop a high-accuracy predictive model for final grades (G3) using only baseline behavioural and socio-economic variables.
- Eliminate data leakage by excluding prior-term grades (G1, G2), enabling genuine early-stage forecasting.

- Identify the factors — such as study time, past failures, and family support — that correlate most strongly with academic success or failure.
- Deploy the trained model into a real-time, user-friendly Streamlit interface so educators can input a student profile and receive an instant risk assessment.

1.4 Scope

The project spans the full machine learning lifecycle: data acquisition and cleaning, exploratory data analysis, feature engineering, model training and hyperparameter optimisation, and final deployment. The model is trained and validated on the UCI Student Performance Dataset, but the underlying feature-selection logic and modelling pipeline are built to generalise — they can be adapted to other institutions as long as comparable data is collected.

1.5 Motivation

At the heart of this project is a fairly simple observation: teachers and counsellors are stretched thin, and manually reviewing every student's situation in detail simply isn't feasible at scale. An automated, data-backed early-warning system doesn't replace human judgement — it multiplies it, directing limited resources like remedial classes and counselling sessions toward the students who need them most, before problems compound.

1.6 Applications

- Early Warning Systems (EWS): automated alerts to academic counsellors when a student's profile trends toward high risk.
- Curriculum Adaptation: helping administrators see whether particular demographics or lifestyle factors are disproportionately affecting outcomes, informing policy changes.
- Adaptive Learning: giving students personalised feedback on their own habits — for instance, the trade-off between time spent socialising and time spent studying — to encourage self-regulation.

1.7 Expected Outcomes

- A validated machine learning pipeline that achieves strong predictive performance without relying on prior-period grades.
- A clear, interpretable analysis of feature importance that explains what actually drives student success.

- A functional, live web application demonstrating the model as a proof of concept for institutional adoption.

Chapter 2: Literature Review

2.1 Existing Approaches

Academic performance prediction has been studied extensively under the umbrella of Educational Data Mining and Learning Analytics. Early work leaned heavily on classical statistical methods — linear and logistic regression — applied to demographic and enrolment data. As richer datasets became available, the field moved toward tree-based ensemble methods (Random Forest, Gradient Boosting) and, more recently, neural network architectures, largely because these models handle non-linear interactions between lifestyle and behavioural variables far better than linear methods.

A large share of published work, however, treats mid-term or continuous-assessment grades as standard predictive features. This produces impressive accuracy figures, but it also means the resulting systems are only useful once a course is already partway through — which is precisely the limitation this project is designed to avoid.

2.2 Research Studies

Several studies using the UCI Student Performance dataset (covering Portuguese secondary school students in Mathematics and Portuguese-language courses) report that including G1 and G2 as predictors noticeably inflates model accuracy compared to excluding them. When these grade features are removed, reported accuracy consistently drops, since the model is left with a genuinely harder problem: forecasting outcomes from behavioural signals alone rather than from a near-final grade already in hand. Exact figures vary across studies depending on algorithm choice and target formulation (regression vs. pass/fail classification), so specific accuracy numbers are not asserted here and should be drawn from the primary sources cited before being presented as fact.

Across this body of research, a consistent set of variables recur as meaningful predictors even without prior grades: number of past class failures, weekly study time, parental education level, school absences, and the quality of family relationships. Ensemble methods such as Random Forest and XGBoost are repeatedly reported as outperforming single classifiers such as decision trees or basic logistic regression on this type of tabular, mixed-type data.

2.3 Research Gap

Two gaps stand out. First, most published models optimise for accuracy using whatever features are available, without deliberately excluding leakage-prone variables — meaning many 'predictive' systems in the literature would be unusable in a genuine early-intervention setting. Second, relatively few studies carry their model through to a deployable, interactive

tool that a non-technical educator could actually use; most stop at offline evaluation metrics reported in a paper or notebook.

This project addresses both gaps directly: by excluding G1 and G2 from the feature set from the outset, and by deploying the final model behind a Streamlit interface intended for direct use by teachers and counsellors rather than data scientists.

Chapter 3: Methodology

3.1 Dataset Description

The project uses the UCI Student Performance Dataset, which records information for secondary school students across two Portuguese schools. The dataset includes 33 attributes spanning demographic details (age, sex, address type), family background (parents' education and occupation, family size, cohabitation status), and lifestyle/behavioural indicators (weekly study time, past class failures, extracurricular activities, alcohol consumption, free time, going out with friends, and health status), alongside the three grade fields G1, G2, and G3.

For this project, G1 and G2 are deliberately dropped from the feature set at the earliest stage of the pipeline, leaving only demographic, family, and behavioural variables as predictors of G3.

3.2 Data Preprocessing

- Handling categorical variables through label encoding and one-hot encoding, depending on whether the category is ordinal or nominal.
- Checking and addressing missing or inconsistent entries, and removing duplicate records.
- Scaling numeric features (e.g., age, absences, study time) using standardisation, which matters particularly for distance- and gradient-based algorithms.
- Converting the continuous G3 score into both a regression target and, separately, a binary pass/fail classification target to support two complementary modelling approaches.
- Splitting the data into training and test sets, with cross-validation used during model selection to guard against overfitting on a relatively small dataset.

3.3 Algorithms Used

A range of algorithms is evaluated to compare interpretability against raw predictive power:

- Logistic / Linear Regression — as an interpretable baseline for classification and regression respectively.
- Decision Tree — for a transparent, rule-based view of how features split students into risk groups.
- Random Forest — an ensemble of decision trees expected to improve robustness and reduce overfitting.

- Gradient Boosting / XGBoost — for their strong track record on structured, tabular datasets of this kind.
- Support Vector Machine (SVM) — evaluated as an alternative non-linear classifier.

Model selection is guided by cross-validated performance, with the final model chosen on the basis of the balance between accuracy, interpretability, and stability across folds.

Chapter 4: Implementation

4.1 Exploratory Data Analysis (EDA)

Exploratory analysis is used to understand the distribution and relationships within the data before any modelling begins. This includes examining the distribution of the target variable (G3) to check for class imbalance in the pass/fail formulation, visualising correlations between behavioural variables (study time, absences, past failures) and final grade, and comparing outcomes across demographic subgroups (school, gender, urban vs. rural address) to surface any structural patterns worth investigating further.

Particular attention is paid to the relationship between 'goout' (time spent socialising) and 'studytime', since these two variables are frequently cited in the literature as having a meaningful, and somewhat intuitive, trade-off effect on performance.

4.2 Feature Engineering

- Encoding binary features (e.g., internet access, romantic relationship status) as 0/1 indicators.
- Creating a composite 'support score' aggregating family support, school support, and paid tuition indicators.
- Binning continuous variables such as age and absences where non-linear threshold effects are suspected.
- Deriving an interaction feature between study time and past failures, since the literature suggests these two variables interact rather than act independently.
- Applying feature importance analysis (via the Random Forest and XGBoost models) to rank variables and, where appropriate, prune ones contributing negligible predictive signal.

4.3 Model Development

Each candidate algorithm is trained on the leakage-free feature set and tuned using grid search or randomised search over key hyperparameters (tree depth, number of estimators, learning rate, regularisation strength, depending on the model). Cross-validation is used throughout to select hyperparameters, and a held-out test set — untouched during tuning — is reserved for final, unbiased evaluation.

The best-performing model is then serialised and wrapped in a Streamlit application, allowing a user to enter a student's demographic and lifestyle attributes through a simple form and

receive an immediate predicted grade or risk category, along with a short explanation of the features driving that particular prediction.

Chapter 5: Results and Discussion

5.1 Performance Metrics

Model performance is assessed using metrics appropriate to each task formulation. For the regression formulation (predicting the numeric G3 score), Root Mean Squared Error (RMSE) and R^2 are used. For the classification formulation (pass/fail), accuracy, precision, recall, F1-score, and the ROC-AUC curve are used, with particular attention paid to recall for the 'at-risk' class — in an early-warning system, missing a genuinely at-risk student is a more costly error than a false alarm.

[Results table pending: to be populated once each candidate model has actually been trained and evaluated on the held-out test set. Report the measured accuracy, F1-score, and ROC-AUC for each of Logistic Regression, Decision Tree, Random Forest, XGBoost, and SVM here, rather than estimated figures.]

5.2 Comparative Analysis

[This section should compare the models once real results are available — for example, which model performed best and why, whether ensemble methods outperformed simpler baselines as expected, and which features the trained models' feature-importance output actually ranked highest. Avoid stating specific performance figures or conclusions until they are backed by the actual experiment output.]

Chapter 6: Conclusion and Future Scope

6.1 Conclusion

This project set out to build a machine learning model capable of predicting student academic performance without relying on prior-term grades — a constraint that makes the problem harder, but also makes the resulting system genuinely usable for early intervention. By training and comparing several algorithms on a leakage-free feature set drawn from demographic, family, and behavioural data, the project identifies which of these factors most meaningfully predict a student's trajectory, and packages the strongest-performing model into a live Streamlit interface for direct use by educators.

The result is not just a model on paper, but a working proof of concept: a tool that could plausibly sit on a teacher's desktop at the start of a term and flag which students may need extra support — before the first exam, not after it.

6.2 Future Scope

- Extending the dataset beyond the two Portuguese schools in the UCI dataset to test generalisability across different institutions and education systems.
- Incorporating longitudinal data — attendance and engagement trends over time — rather than a single static snapshot of behavioural attributes.
- Exploring fairness and bias auditing to ensure the model does not disproportionately flag students from particular demographic groups.
- Integrating the tool with existing school learning management systems for automatic data ingestion rather than manual entry.
- Adding explainability tooling (e.g., SHAP values) directly into the Streamlit interface so educators can see, in plain language, why a given student was flagged.

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